Network design practice 01-1

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1 Overview

This practice simulates basic SDN structure.

2 Assignment 1

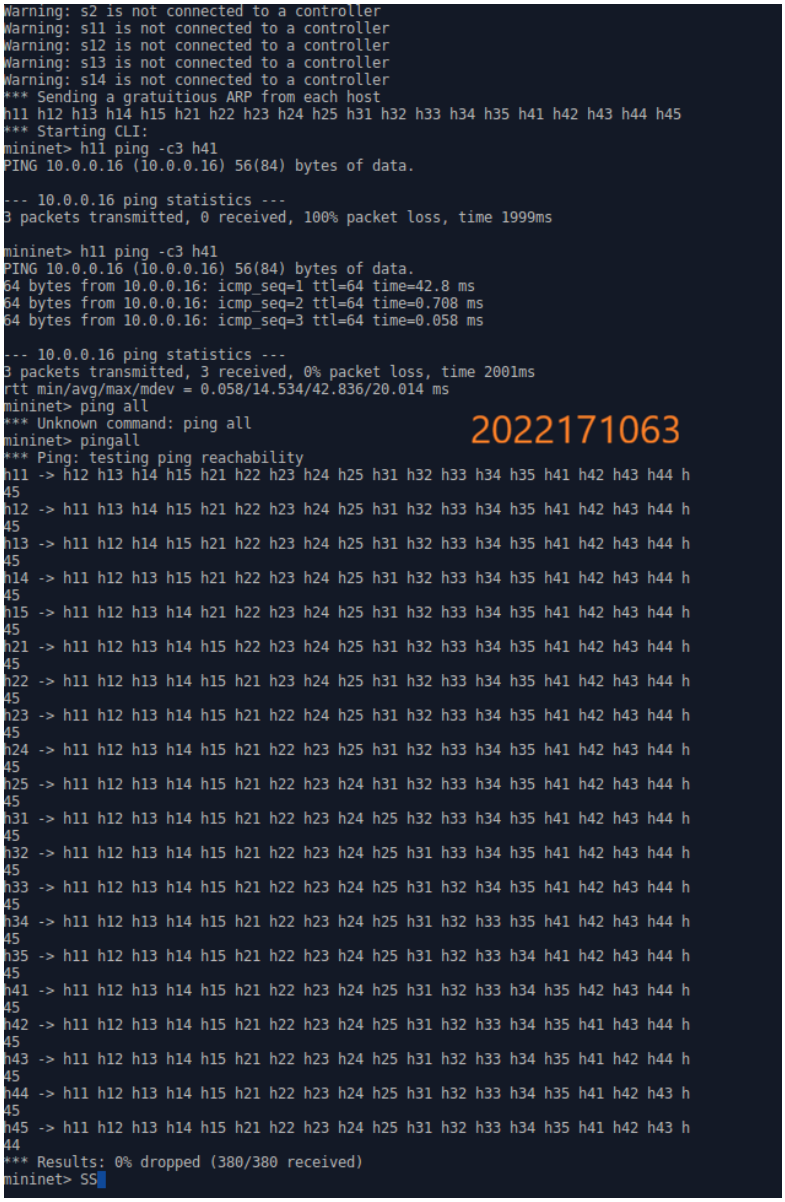
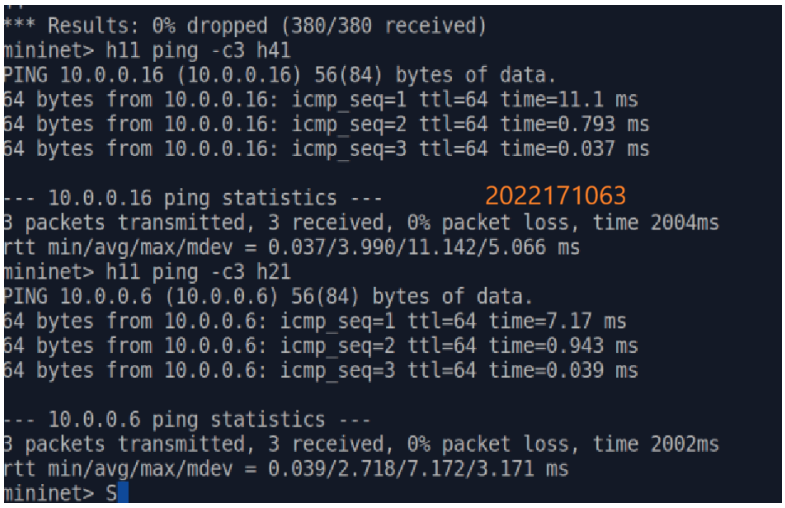


Figure 1: result of pingall

pingall is a command to observe the reachability among all hosts. Starting from h11 it takes a full round trip to h45, sending ping to reach each and every other hosts on the list. We observe all 380 packets are sent and received with no errors, therefore the reachability is up and functional.

3 Assignment 2



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*** Results: 0% dropped (380/380 received)
mininet> h11 ping -c3 h41
PING 10.0.0.16 (10.0.0.16) 56(84) bytes of data.
64 bytes from 10.0.0.16: icmp_seq=1 ttl=64 time=11.1 ms
64 bytes from 10.0.0.16: icmp_seq=2 ttl=64 time=0.793 ms
64 bytes from 10.0.0.16: icmp_seq=3 ttl=64 time=0.037 ms

--- 10.0.0.16 ping statistics ---        2022171063
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 0.037/3.990/11.142/5.066 ms
mininet> h11 ping -c3 h21
PING 10.0.0.6 (10.0.0.6) 56(84) bytes of data.
64 bytes from 10.0.0.6: icmp_seq=1 ttl=64 time=7.17 ms
64 bytes from 10.0.0.6: icmp_seq=2 ttl=64 time=0.943 ms
64 bytes from 10.0.0.6: icmp_seq=3 ttl=64 time=0.039 ms

--- 10.0.0.6 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 0.039/2.718/7.172/3.171 ms
mininet> S
```

Figure 2: ping test

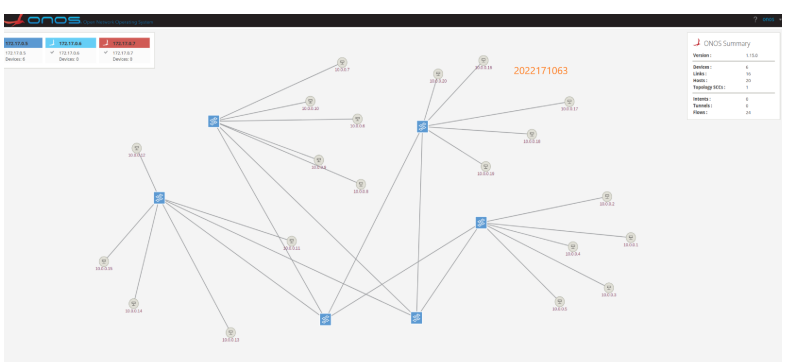
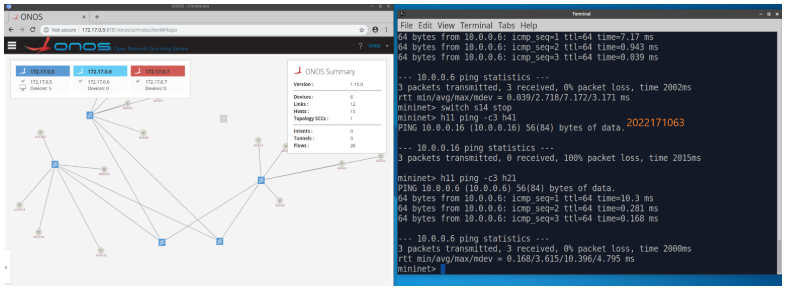


Figure 3: topology on ONOS GUI

Before stop, ping test from h11 to h41, h11 to h21 is successful with no errors.



After stop, the switch s14 is colored grey indicating it is inactive. Packets bound for h41 is never reaching because h41 is only connected to switch s14 which is no longer functional. Packet bound for h21 is received normally since its path from h11 is unaffected.